

DSA Tutorial - 2

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Topic: Binary Search

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INPUT :

```
#include <stdio.h>
```

```
int binarySearch(int arr[], int low, int high, int target) {  
    if (low > high) {  
        return -1;  
    }
```

```
    int mid = low + (high - low) / 2;
```

```
    if (arr[mid] == target) {  
        return mid;  
    }
```

```
    if (arr[mid] > target) {  
        return binarySearch(arr, low, mid - 1, target);  
    }
```

```
    return binarySearch(arr, mid + 1, high, target);  
}
```

```
int main() {  
    int n, target;
```

```
    printf("Enter the number of elements: ");  
    scanf("%d", &n);
```

```
    int arr[n];
```

```
    printf("Enter %d sorted elements:\n", n);  
    for (int i = 0; i < n; i++) {  
        scanf("%d", &arr[i]);  
    }
```

```
printf("Enter the element to search: ");
scanf("%d", &target);

int result = binarySearch(arr, 0, n - 1, target);

if (result != -1) {
    printf("Element found at index: %d\n", result);
} else {
    printf("Element not found\n");
}

return 0;
}
```

OUTPUT :

```
Output

Enter the number of elements: 5
Enter 5 sorted elements:
4 6 7 8 9
Enter the element to search: 7
Element found at index: 2

=== Code Execution Successful ===
```